

Electrical Principles and Practice

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1 Fundamentals of Electricity

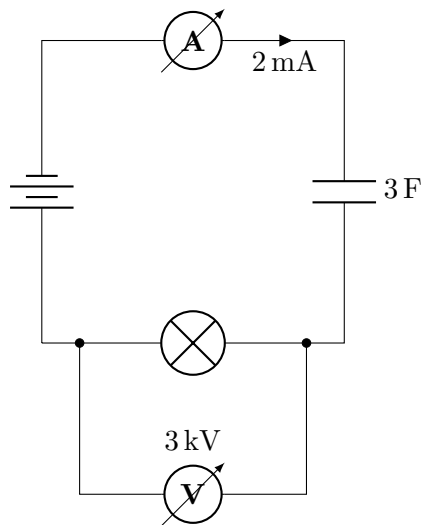
1.1 Basic quantities

$$V = \frac{W}{Q}, \quad I = \frac{dQ}{dt}, \quad P = VI. \quad (1)$$

Ohm's law is

$$V = IR. \quad (2)$$

1.2 Example circuit



2 Thevenin Equivalent Circuits

Any linear two-terminal network can be replaced by a Thevenin voltage source V_{th} in series with a resistance R_{th} :

$$V_{\text{th}} = V_{\text{oc}}, \quad R_{\text{th}} = \frac{V_{\text{oc}}}{I_{\text{sc}}}. \quad (3)$$

3 Node Voltage Analysis

Choose a reference node, assign a voltage to every remaining node, and apply Kirchhoff's current law:

$$\sum_k \frac{V_n - V_k}{R_k} = I_{\text{injected}}. \quad (4)$$

4 Capacitors

For a capacitor,

$$q = Cv, \quad i = C \frac{dv}{dt}, \quad w = \frac{1}{2} Cv^2. \quad (5)$$

For series and parallel combinations,

$$\frac{1}{C_{\text{series}}} = \sum_i \frac{1}{C_i}, \quad C_{\text{parallel}} = \sum_i C_i. \quad (6)$$

5 Capacitors and Inductors

An inductor satisfies

$$v = L \frac{di}{dt}, \quad w = \frac{1}{2} Li^2. \quad (7)$$

The impedance of ideal reactive components at angular frequency ω is

$$Z_C = \frac{1}{j\omega C}, \quad Z_L = j\omega L. \quad (8)$$

6 AC Circuits and Phasors

For sinusoidal steady-state analysis, represent signals using phasors:

$$v(t) = V_m \cos(\omega t + \phi) \quad \longleftrightarrow \quad \tilde{V} = V_{\text{rms}} \angle \phi. \quad (9)$$

Impedances can be combined using the same series and parallel rules as resistances. The average power is

$$P = V_{\text{rms}} I_{\text{rms}} \cos \phi. \quad (10)$$